International Institute of Information Technology, Hyderabad
MA4.101-Real Analysis (Monsoon-2026)

Practice Problems with Solutions

Chapter 5: The real number system

Definitions and facts assumed

The solutions below are self-contained: every result used in a proof is stated where it is used. The following definitions and facts are assumed throughout and are not reproved.

Definitions.

(D1) For a real $x \geq 0$ and an integer $n \geq 1$, $x^{1/n} := \sup\{\, y \in \mathbb{R} : y \geq 0 \text{ and } y^n \leq x \,\}$.

(D2) For a real $x > 0$ and a rational $q = a/b$ with $a \in \mathbb{Z}$ and $b \geq 1$ an integer, $x^q := \left(x^{1/b}\right)^a$.

Facts.

(F1) *(Least upper bound property.)* Every non-empty subset of $\mathbb{R}$ which has an upper bound has exactly one least upper bound.

(F2) For a real $t \geq 0$ and integers $n \geq 1$ the usual laws of exponents hold, and $t \mapsto t^n$ is strictly increasing on $\{t \geq 0\}$: if $0 \leq s < t$ then $s^n < t^n$.

(F3) For $x \geq 0$ and $n \geq 1$, the number $x^{1/n}$ satisfies $\left(x^{1/n}\right)^n = x$; conversely, if $y \geq 0$ and $y^n = x$ then $y = x^{1/n}$. Moreover $x^{1/n} > 0$ whenever $x > 0$.

(F4) Definition (D2) does not depend on the representation of q: if $a/b = a'/b'$ with $b, b' \geq 1$, then $(x^{1/b})^a = (x^{1/b'})^{a'}$.

Question (1). Let $x \geq 0$ be a non-negative real, and let $n \geq 1$ be a positive integer. Show that the set $E := \{\, y \in \mathbb{R} : y \geq 0 \text{ and } y^n \leq x \,\}$ is non-empty and is also bounded above. In particular, deduce that $x^{1/n}$ is a real number.

Non-empty. We have $0 \geq 0$ and $0^n = 0 \leq x$, so $0 \in E$.

Bounded above. We claim $x + 1$ is an upper bound for E.

First note that $t^n \geq t$ whenever $t \geq 1$. This is proved by induction on n: at $n = 1$ it reads $t \geq t$; and if $t^n \geq t$ then $t^{n+1} = t^n \cdot t \geq t \cdot 1 = t$, using $t \geq 1$ and $t^n \geq t \geq 1 > 0$.

Now let $y > x + 1$. Since $x \geq 0$ we get $y > 1$, so by the previous paragraph $y^n \geq y > x + 1 > x$, and therefore $y \notin E$. Equivalently, every $y \in E$ satisfies $y \leq x + 1$.

Conclusion. E is a non-empty subset of $\mathbb{R}$ with an upper bound, so by (F1) it has a least upper bound, which is a real number. By definition (D1) this least upper bound is $x^{1/n}$.

Question (2). Let $x, y \geq 0$ be non-negative reals, and let $n, m \geq 1$ be positive integers. Prove the following.

(a) We have $x > y$ if and only if $x^{1/n} > y^{1/n}$.

(b) If $x > 1$, then $x^{1/k}$ is a decreasing function of k. If $x < 1$, then $x^{1/k}$ is an increasing function of k. If $x = 1$, then $x^{1/k} = 1$ for all k.

(c) We have $(x^{1/n})^{1/m} = x^{1/nm}$.

By Question (1) the number $x^{1/n}$ is a real number for every $x \geq 0$ and every $n \geq 1$, and by (F3) it is the unique non-negative real whose nth power is x. We use this throughout.

(a) Write $a := x^{1/n}$ and $b := y^{1/n}$, so that $a, b \geq 0$ and, by (F3), $a^n = x$ and $b^n = y$.

If $a > b$, then $a^n > b^n$ by (F2), that is, $x > y$.

Conversely suppose $x > y$. By trichotomy exactly one of $a > b$, $a = b$, $a < b$ holds. If $a = b$ then $x = a^n = b^n = y$, impossible. If $a < b$ then $x = a^n < b^n = y$ by (F2), also impossible. Hence $a > b$.

(b) First, $1^{1/k} = 1$ for every $k \geq 1$: the number 1 is non-negative and satisfies $1^k = 1$, so the uniqueness clause of (F3) identifies it as $1^{1/k}$. This settles the case $x = 1$.

Now fix $k \geq 1$ and write $a := x^{1/k}$ and $b := x^{1/(k+1)}$, so that $a, b \geq 0$ and, by (F3), $a^k = x = b^{k+1}$.

Case $x > 1$. Applying (a) to the pair x and 1, once with exponent k and once

with exponent $k+1$, gives $a > 1^{1/k} = 1$ and $b > 1$. Suppose $b \geq a$. Then by (F2),

$$b^{k+1} \;\geq\; a^{k+1} \;=\; a^k \cdot a \;>\; a^k \;=\; b^{k+1},$$

using $a > 1$ and $a^k > 0$; this is impossible. Hence $b < a$, that is, $x^{1/(k+1)} < x^{1/k}$, so $x^{1/k}$ is a decreasing function of k.

Case $0 < x < 1$. Applying (a) to the pair 1 and x gives $1 = 1^{1/k} > a$ and likewise $1 > b$; and $a > 0$ by (F3), since $x > 0$. Suppose $b \leq a$. Then

$$b^{k+1} \;\leq\; a^{k+1} \;=\; a^k \cdot a \;<\; a^k \;=\; b^{k+1},$$

using $0 < a < 1$; impossible. Hence $b > a$, so $x^{1/k}$ is an increasing function of k.

(For $x = 0$ one has $0^{1/k} = 0$ for every k, so the second statement is to be read for $x > 0$.)

(c) Put $c := (x^{1/n})^{1/m}$, so that $c \geq 0$. By (F3) applied first with exponent m and then with exponent n,

$$c^m = x^{1/n}, \qquad \text{hence} \qquad c^{nm} = (c^m)^n = (x^{1/n})^n = x.$$

So c is a non-negative real whose nmth power is x, and the uniqueness clause of (F3), applied with exponent nm, gives $c = x^{1/nm}$.

Question (3). Let $x, y > 0$ be positive reals, and let q, r be rationals. Prove the following.

(a) x^q is a positive real.

(b) $x^{q+r} = x^q x^r$ and $(x^q)^r = x^{qr}$.

(c) If $x > 1$, then $x^q > x^r$ if and only if $q > r$. If $x < 1$, then $x^q > x^r$ if and only if $q < r$.

Throughout write $q = a/b$ and $r = c/d$ with $a, c \in \mathbb{Z}$ and $b, d \geq 1$ integers, so that $x^q = (x^{1/b})^a$ and $x^r = (x^{1/d})^c$ by (D2); by (F4) the value does not depend on the representation chosen.

A reduction to a common root. Put $v := x^{1/bd}$, which is positive by (F3). We claim

$$x^q = v^{\,ad} \qquad \text{and} \qquad x^r = v^{\,bc}.$$

Indeed $(v^d)^b = v^{\,db} = x$ by (F3), and $v^d \geq 0$, so the uniqueness clause of (F3) gives $v^d = x^{1/b}$. Hence $x^q = (x^{1/b})^a = (v^d)^a = v^{\,ad}$. The second identity is the

same computation with b and d exchanged.

(a) By (F3), $x^{1/b} > 0$. A positive real raised to an integer power is positive: for a positive exponent this is a product of positive numbers, for exponent 0 the value is 1, and for a negative exponent it is the reciprocal of a positive number. Hence $x^q = (x^{1/b})^a > 0$.

(b) *The sum law.* We have $q + r = (ad + bc)/(bd)$, so by (D2) and the reduction above,

$$x^{q+r} \;=\; \left(x^{1/bd}\right)^{ad+bc} \;=\; v^{\,ad+bc} \;=\; v^{\,ad}\, v^{\,bc} \;=\; x^q\, x^r,$$

the third equality being the law of integer exponents in (F2).

The power law. Since $x^q = v^{\,ad}$ and $v > 0$, we first identify $(x^q)^{1/d}$. The number v^a is positive and $(v^a)^d = v^{\,ad} = x^q$, so the uniqueness clause of (F3) gives $(x^q)^{1/d} = v^a$. Hence, by (D2),

$$(x^q)^r \;=\; \left((x^q)^{1/d}\right)^c \;=\; (v^a)^c \;=\; v^{\,ac}.$$

On the other hand $qr = ac/(bd)$, so $x^{qr} = (x^{1/bd})^{ac} = v^{\,ac}$. The two agree.

(c) We note two auxiliary facts first.

Reciprocals. For every rational s we have $x^{-s} = 1/x^s$. Indeed, write $s = e/f$ with $e \in \mathbb{Z}$ and $f \geq 1$; then $-s = (-e)/f$, so by (D2) and the law of integer exponents in (F2),

$$x^{-s} \;=\; \left(x^{1/f}\right)^{-e} \;=\; \frac{1}{(x^{1/f})^e} \;=\; \frac{1}{x^s},$$

which is legitimate because $x^{1/f} > 0$ by (F3).

Powers bigger than one. Suppose $x > 1$, and let $s = e/f$ be rational with $f \geq 1$. Applying Question (2)(a) to the pair x and 1 gives $u := x^{1/f} > 1^{1/f} = 1$. For an integer $e \geq 1$ we then have $u^e > 1$, being a product of numbers exceeding 1; for $e = 0$ we have $u^0 = 1$; and for $e \leq -1$ we have $u^e = 1/u^{-e} < 1$. So $x^s = u^e > 1$ exactly when $e > 0$, that is, exactly when $s > 0$.

The case $x > 1$. By (b) and the reciprocal rule,

$$\frac{x^q}{x^r} \;=\; x^q\, x^{-r} \;=\; x^{\,q-r}.$$

Since $x^r > 0$ by (a), we have $x^q > x^r$ if and only if $x^{\,q-r} > 1$, which by the previous paragraph holds if and only if $q - r > 0$, that is, if and only if $q > r$.

The case $0 < x < 1$. Here the second auxiliary fact reverses: applying Question (2)(a) to the pair 1 and x gives $0 < u < 1$, so $u^e > 1$ exactly when $e < 0$, and hence $x^s > 1$ exactly when $s < 0$. The same computation as above therefore gives $x^q > x^r$ if and only if $q - r < 0$, that is, if and only if $q < r$.